Sliding, swinging and rolling

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Learning target: Choose the energy terms that match the type of motion.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 7; 4 course days later.*
+240 J; the object gains gravitational potential energy.

02. Explain a worked example (worked)
Initial U 20; final rotational K 8 and translational K 12; all must sum consistently.

03. Look and explain (model)
Some energy is rotational kinetic energy, leaving less for center translation.

04. Build the solution (practice)
15 J.

05. Use the idea (reason)
Tension does zero work; gravity changes K.

06. Choose a method (mixed) *Review: Unit 3, Day 13; 12 course days later.*
Shift 24 m. Assumes a constant added transverse velocity and a straight-line comparison.

07. Exit check (exit)
Wheelspins, so it also has rotational kinetic energy.

08. STRETCH: Apply it another way (stretch)
Rolling stores part in rotation; sliding can put more into translation. Models and rotational properties matter.

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01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 7; 4 course days later.*
ΔU=-240 J; gravity does +240 J work.

02. Explain a worked example (worked)
Initial U 20; final rotational K 8 and translational K 12; all must sum consistently.

03. Represent and interpret (model)
Some energy is rotational kinetic energy, leaving less for center translation.

04. Solve and justify (practice)
mgh=.75 mv²; v=√(4 gh/3)=√20=4.47 m/s. Point slider would√30=5.48 m/s.

05. Analyze the reasoning (reason)
Tension does zero work; gravity changes K.

06. Choose a method (mixed) *Review: Unit 3, Day 13; 12 course days later.*
Shift multiplier 6/4≈1.5 because shift=(J/M) t. Assumes constant added transverse velocity.

07. Exit check (exit)
4.0 m/s.

08. STRETCH: Apply it another way (stretch)
mgh=½mv²+¼mv²=¾mv²; v=√(4 gh/3)=√20≈4.47 m/s.